

Contact

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(LinkedIn)

Top Skills

Nuclear Medical Imaging

telecom lasers

THz spectroscopy

Certifications

XCS221 - Artificial Intelligence:
Principles and Techniques

Don Burdette

Principal Engineer

San Francisco Bay Area

Summary

Experienced Principal Engineer with a Ph.D. in Physics, specializing in advanced test technologies and optical systems, with a proven track record in system development and successful project leadership at Analog Devices, Maxim Integrated, and Traycer.

Experience

Analog Devices

Principal Engineer - Advanced Test Technology

January 2022 - Present (4 years 8 months)

San Jose, CA

Major Projects (Completed)

- Developed custom vision inspection and handler system for products with non-standard dimensions.

Current Projects

- Developing constraint-satisfaction software to determine ideal probe-head geometries and optimized scanning patterns. The goal is to minimizing wafer-level test times.

Maxim Integrated

Optical Physicist - Advanced Test Technology

October 2017 - January 2022 (4 years 4 months)

San Jose, CA

Major Projects:

- Developed calibration hardware, methods, and software for optical product lines.
- Developed power delivery network (PDN) layout principles and simulation automation software for quality control test PCBs.

Traycer, Inc

Senior Scientist

2009 - September 2017 (8 years)

Newark, CA

- Led development of THz systems for industrial and medical applications.
- Designed and programmed system software for control and data acquisition.
- Secured a \$110,000 SBIR grant for military aircraft inspection technology.

The Ohio State University

7 years

Graduate Research Assistant

2005 - 2009 (4 years)

The Ohio State University

Responsible for the development and implementation of the hardware and software for nuclear medicine based imaging systems.

- Designed and constructed a Positron Emission Tomography (PET) nuclear medical imaging device based on novel silicon pad detector technology capable of operating in high magnetic field environments.
- Wrote maximum likelihood estimation image processing routines to remove systematic sources of noise in order to achieve state-of-the-art image resolution for high-energy positron emitting radioactive tracers.

Graduate Teaching Assistant

2002 - 2005 (3 years)

The Ohio State University

Lead lecture and lab based physics recitation courses for both the general and engineering sequences. Subjects taught included: Newtonian mechanics, electricity and magnetism, circuit theory, optics, modern physics, and nuclear physics.

Oak Ridge National Laboratory

Intern

2000 - 2001 (1 year)

Oak Ridge TN

Studied the degradation of super conductive cable insulation and developed a technique for predicting both the type of failure and the remaining lifetime of the device. This work involved electromagnetic simulations based on data collected from dielectric samples injected with various failure inducing defects.

Education

The Ohio State University

Ph.D., Physics · (2002 - 2009)

Indiana University of Pennsylvania
Bachelor's Degree, Physics · (1998 - 2002)